Feasibility Assessment

Project AutoEat — Layer 3 physical prep robot (v1)

Independent technical / financial / timeline review · prepared for the AutoEat team

> **Scope of what was reviewed.** Only the handoff brief was present in the working folder. The referenced supporting files (01_Plan_and_Timeline, 02_Architecture, 03_Layer1_Planner, 05_Hardware, 04_Moral_Implications, and AutoEat_Cost_Sheet.xlsx) were not available, so their internal numbers could not be cross-read. This assessment is built from the brief plus current external sources (hardware specs, prices, method results), and every point that would have depended on an unread file is flagged as such.

1. Executive summary

AutoEat v1, as scoped, is a sound and buildable student project — but only because the scope has already been cut down to the one shape that works. A ~$200–400 SO-101 arm assembling one pre-portioned cold dish, learned with ACT on 50-plus teleoperated demonstrations, sits squarely inside what the low-cost-manipulation community has already demonstrated many times. Expect roughly 60–80% single-task success after real tuning, not the 99% people associate with "a robot that makes breakfast." The engineering risk is mostly known-how; the genuine open risk is narrow and nameable — reliable spoon-to-cup scooping and pouring of granular/viscous food (a contact-rich, deformable-material problem), plus the brittleness of a fixed-camera policy the moment anything in the scene moves. The budget is real for parts but optimistic as a true project cost; plan for 1.5–2x. Twelve months is enough only if the team treats data collection and iteration — not assembly — as the main event.

2. Overall feasibility rating

Verdict: **Feasible, with caveats**

Confidence: **Moderate-to-high on v1 as scoped; low on the broader "food robot" vision**

The rating applies to the narrow v1 target (one cold-assembly dish, pre-portioned inputs, attended operation). It is **not** a verdict on the three-layer vision or on any version that cooks, cuts, or handles heavy cookware — those are separate, harder projects (see the heavy-lifting note below, which the brief asked to be kept open).

Note on heavy-object handling (door kept open, per request)

The brief's "Path A only / nothing over 400 g" decision is the right call **for v1**, but it does not have to be permanent, and it is worth stating exactly where the ceiling is so the door can stay open honestly.

- **What the hardware actually allows.** Each STS3215 servo is rated ~30 kg·cm stall torque at 12V (the SO-101 spec) — but stall torque is not usable payload. Because torque is force times lever arm, a ~500 mm arm extended horizontally converts that rating into only a few hundred grams of dependable end-effector payload; independent analyses put the practical envelope at a few hundred grams up to ~1 kg depending on how extended the arm is and how the load sits.
- **So the door that is realistically open:** light-to-moderate objects (a filled cup, a small yogurt tub, a plastic bowl with food in it) handled close to the base, at low speed, is plausible on this exact arm — that is more than 400 g and still fine. A full 1 kg reach-out lift is at the ragged edge and will overheat servos and flex the PLA frame under repetition.
- **The door that is effectively closed on this arm:** a filled pot or pan (easily 1.5–3 kg), or any heavy lift at reach. That is not a tuning problem; it is a torque and structural-stiffness problem. Keeping it open means a different, stronger arm later (higher-torque servos or a larger platform) — a hardware swap, not a v1 setting. The LeRobot software and your collected data would largely carry over.

Recommendation: keep v1 at pre-portioned cold assembly, but do not write "400 g" into the design as a hard law. Frame it as "light objects near the base for now," and design the workspace so a future stronger arm could drop in. That preserves the heavy-handling ambition without letting it sink v1.

3. Dimension-by-dimension assessment

3.1 Technical feasibility — Feasible-with-caveats (highest-priority dimension)

Can a ~500 g-payload SO-101 manipulate a spoon and light cups for one cold dish? Yes, in principle, and the community has done close analogues. But "pick up a rigid cube and place it" (the standard 70%-success demo) is meaningfully easier than "scoop oats, level the spoon, carry without spilling, and pour yogurt." The hard part is not the arm reaching positions; it is the contact-rich, granular/viscous, partially deformable nature of food.

Realistic reliability and demo count. ACT's own documentation says ~50 demonstrations is often enough for a single-variation task, trained ~100k steps in a few GPU-hours. A widely-cited SO-101 ACT pick-and-place write-up reached 70% success from 50 episodes; other LeRobot ACT tasks land in the 70–90% band with careful data. Translate that honestly:

- Rigid pick-and-place sub-steps (grab spoon, move cup): expect 80–90% per step after tuning.
- Scooping/pouring food: expect lower and noisier — 50–70% per step is a fair early expectation — and full-dish success is the **product** of the steps, so a 4-step chain at 80% each is only ~40% end-to-end. This chaining math is the single most under-appreciated risk in the plan.
- Getting from there to a clean demo will likely need 100–300 demonstrations, task-specific staging, and possibly splitting the dish into sub-policies rather than one monolithic policy.

Perception (single fixed scene camera + wrist camera): adequate, with one hard rule. This exact two-camera setup is the LeRobot default and is proven. The catch is already noted in the brief and is correct: the scene camera must be rigidly fixed and the lighting/scene layout stable. ACT policies are notoriously brittle to camera shifts and distribution changes — bump the camera and the policy silently degrades. RGB-only (no depth) is fine for this scale; RealSense is a genuine "optional."

Honest hardest sub-problem. Granular/viscous scooping-and-pouring with spill tolerance, delivered reliably from a fixed-camera ACT policy on a low-stiffness arm. Verdict: this is **engineering risk with a research-risk tail**. Pick-and-place is solved-pattern; food-consistency manipulation at high reliability on this hardware class is demonstrated in pieces but not a turnkey recipe. Budget iteration time for it and it is very likely achievable at demo quality; assume 95%+ reliability and it becomes research risk.

3.2 Financial feasibility — Feasible on parts, optimistic as project cost

The ~$269 single-arm / ~$393 leader-follower figures are believable for the arm itself: SO-101 kits and DIY builds are widely quoted from ~$150 (pure DIY) to ~$199+ (kit), and STS3215 servos run ~$13–15 each. The numbers are not fantasy. What the headline budget tends to omit:

- Filament and failed prints (the SO-101 is a large multi-part print; first-timers waste 1–3 spools) — $40–80.
- Spare servos. STS3215s are the most likely thing to burn out under repeated near-limit torque or a crash. Buy at least one or two spares — $15–45.
- Cameras, if not scavenged — 2x decent USB webcams, $30–70; a good overhead mount/rig, some more.
- Incidentals: USB hub, cabling, clamps, a sturdy work surface, ingredient cups, cleanable mat, small hand tools — $40–100.
- Shipping/import on servos and boards — $15–40.

True expected spend with contingency: plan ~$400–550 for the single-arm build and ~$550–750 for a leader-follower pair, i.e. roughly 1.5–2x the headline. Still a very cheap capstone. Applying the brief's own 20–30% contingency to a corrected base (not the headline) is the right way to book it.

3.3 Timeline feasibility — Feasible but front-loaded with hidden slippage

Twelve months part-time is enough **if** the team understands where the time actually goes. Assembly is the easy, fast part (build is quoted at 2–4 hours; realistically a week or two with printing, calibration, and first-time debugging). The schedule is eaten by three things:

- **Data collection ergonomics.** Collecting 100–300 clean teleop demonstrations is slow, repetitive human labor, and early datasets are usually thrown away once the team learns what "good" looks like.
- **The train–evaluate–fix loop.** Real-robot evaluation is manual, subjective, and time-consuming; you cannot trust validation loss, so every iteration costs bench time.
- **The reliability grind.** Going from "works in a demo" (60%) to "works when the professor walks in" (repeatable) is the long tail and is where capstones slip.

Where it slips: months 1–3 will feel productive (arm builds, teleop works, first policy moves). The slump is months 6–9, when the food-manipulation reliability plateau hits. Protect that window and cut dish complexity before cutting the schedule.

3.4 Team / skill feasibility — Plausible for a fourth-year team

Skills required, and the honest difficulty for a strong senior team:

Skill	Needed for	Difficulty for a 4th-year team
Mechanical assembly / 3D printing	Building and maintaining the arm	Low — guided, plug-and-play, well-documented
Embedded / servo bring-up	Calibration, serial comms, drivers	Low-moderate — mostly following LeRobot setup
Python + ML tooling	Running LeRobot, training ACT	Moderate — the stack is turnkey, the debugging is not
Data collection discipline	Good demonstrations	Moderate — underrated; quality is a learned skill
Systems debugging / patience	The reliability grind	Steepest curve — this is where teams stall

No single skill here is out of reach for a data-science/robotics senior team with a faculty advisor. The steepest curve is not a technology; it is the **empirical debugging loop** — diagnosing why a policy fails and knowing whether the fix is more data, better data, camera stability, or task redesign. A robotics professor's guidance materially de-risks this.

3.5 Safety — Envelope is adequate for v1, with standard cautions

The "no heat, no blades, nothing over 400 g" envelope is genuinely the right safety posture and makes v1 benign. Remaining, manageable concerns:

- **Pinch/crush points.** Even a low-payload arm can pinch fingers at joints and the gripper. Keep hands clear during autonomous runs; add an easy physical e-stop / power cutoff.
- **Food hygiene.** PLA is porous and not food-safe long-term, and printed parts harbor bacteria. Anything contacting food (spoon, cups) should be food-grade and cleanable, ideally not the printed parts themselves. This is a real issue precisely because the output is eaten.

- **Unattended operation: don't, in v1.** Treat the robot as attended-only. A low-stiffness arm can knock a cup, mislocate, or fling food; unattended runs also compound the hygiene risk. Attended operation keeps the whole envelope safe.

3.6 IP / institutional — Flag, don't resolve

If university printers, lab hardware, GPU clusters, or faculty inventive contribution are used, institutional IP policy may attach to what the team produces. This is routine for capstones and usually resolvable, but it should be clarified in writing **before** the project acquires any commercial ambition. Action: ask the sponsoring department early who owns project outputs and whether the open-source (Apache-2.0-style) nature of LeRobot/SO-ARM simplifies things. Not a technical blocker; a paperwork item to not discover late.

4. The single biggest risk

Reliability collapse from step-chaining on contact-rich food manipulation. The project's success is the product of several imperfect sub-steps, and food (granular oats, viscous yogurt) is exactly the material class where per-step success drops. A plausible 4-step chain at 70–80% per step yields ~30–40% end-to-end — which reads as "it barely works" even though every piece "works."

What most reduces it:

- Pick the dish to minimize step count and eliminate pouring where possible (favor scoop-and-drop of forgiving solids over precise pours).
- Consider sub-policies per step with scripted transitions, rather than one end-to-end policy, so failures are isolated and debuggable.
- Over-collect demonstrations on the worst step specifically, and lock the camera and scene like it is a lab instrument.
- Define "success" concretely and early (e.g. "edible bowl assembled within tolerance in N of 10 trials") so the team optimizes the right number.

5. Assumption audit

Assumption (Section 5)	Status	Basis / note
Makerspace printing is free/available to this team	Unverified	Team/institution-specific; the missing 05_Hardware and plan docs would state this. Confirm printer access, bed size for large SO-101 parts, and filament supply directly.
SO-101 ~500 g payload & ~500 mm reach cover the object set	Verified with qualification	Servos are ~30 kg·cm stall @12V; usable payload is a few hundred grams to ~1 kg near-base. Adequate for spoon + light cups; NOT for heavy cookware. Holds for v1 object set.
Free-tier cloud compute suffices to train the policy	Partly verified / caution	ACT trains in a few GPU-hours for ~100k steps, which fits a single session in principle — but Colab free tier has session caps, timeouts, and no guaranteed GPU. Workable with checkpointing to Drive; a university cluster is the safer primary. Don't rely on free Colab alone for many iterations.
Kinesthetic teaching yields usable demos (skipping the	Plausible but weaker —	Kinesthetic/hand-guided teaching works and saves the leader-arm cost, but tends to produce

Assumption (Section 5)	Status	Basis / note
leader arm)	verify	jerkier, noisier trajectories than a smooth leader-follower teleop pair, and ACT is sensitive to demo quality. For a contact-rich food task, the leader-follower pair is worth the extra ~$120. Treat single-arm as a cost floor, not the recommended path.
A cold-assembly dish exists that is nutritious AND simple enough	Verified	Yes — overnight oats / assembled yogurt-grain bowl. Nutritious, no heat, forgiving of imprecision, minimal step count. See Section 7.

Note: three of the five assumptions ultimately depend on documents that were not in the folder (printing access, exact BOM, plan-doc compute claims). Re-check those against 05_Hardware and the cost sheet once available.

6. Recommended go / no-go and next concrete step

Recommendation: **GO — on the scoped v1, with the leader-follower build, not the single-arm cost-floor build**

First thing to buy/build: Order one SO-101 follower kit plus the leader arm (the ~$393 pair, corrected to ~$550–750 all-in), and in parallel start printing and confirm makerspace access before spending on electronics. The very first milestone is not the food task — it is: assemble the arm, get LeRobot teleoperation working, and collect + train a trivial rigid pick-and-place (grab the spoon, move it) to 80%+. That single loop proves the entire toolchain end-to-end and de-risks everything downstream for well under $100 of additional effort. Only after that loop closes should the team touch food.

What would flip this to no-go: if makerspace printing/GPU access turns out unavailable (pushing true cost and complexity up sharply), or if the team cannot commit the sustained mid-project bench time the reliability grind demands. Neither is a technology failure; both are resourcing failures — so confirm resources before committing the year.

7. Recommended first-target dish

Overnight oats (single pre-portioned bowl). Best first target because every hard edge is filed off: no heat, no cutting, forgiving of imprecise scoops, few steps (drop oats, add yogurt/milk cup, add toppings, stir), and the failure mode is a slightly messy but still-edible bowl rather than anything dangerous. An assembled yogurt-grain bowl is the near-equivalent fallback. Avoid anything requiring precise liquid pouring or level measuring as the first dish — earn that later.

8. Sources

- SO-ARM100/101 official repository and specs — github.com/TheRobotStudio/SO-ARM100 (6-DOF SO-101, STS3215 servos, open-source).
- LeRobot SO-101 documentation — huggingface.co/docs/lerobot/so101 (follower uses 6x STS3215, 1/345 gearing; assembly and setup).
- Feetech STS3215 servo spec — ~30 kg·cm stall torque @12V (19.5 kg·cm @7.4V), 12-bit magnetic encoder, position/load/temperature feedback (Waveshare SO-ARM100/101 wiki; Feetech listing).
- Practical payload analysis — usable end-effector payload a few hundred grams to ~1 kg depending on config, well below stall-torque rating due to gear losses and moment arms (RobotStudio SO-ARM100 analysis, refft.com).

- ACT policy data-efficiency — ACT docs: ~50 demonstrations often sufficient for a single-variation task, ~100k training steps, a few GPU-hours (huggingface.co/docs/lerobot/main/en/act; LeRobot DeepWiki ACT page).
- ACT real-world success rate on SO-101 — ~70% on a 50-episode pick-and-place task (Karkada, Medium, 2025); comparable LeRobot ACT tasks in the 70–90% band (arXiv 2409.17435; 2503.01543).
- Evaluation difficulty / policy brittleness — field report on LeRobot imitation learning; real-world human evaluation is labor-intensive and subjective (ml6.eu, 2025).
- Build time and cost — SO-101 assembly ~2–4 hours; DIY build from ~$150, kits from ~$199 (roboticscenter.ai/hardware/so-101; SVRC).
- Kit/electronics pricing reference — SO-101 follower electronics + frame kits, STS3215 servos ~$13–15 each (vendor listings, 2026).

Could not verify (flagged): makerspace printer availability and free-tier GPU quota for THIS team; the exact bill of materials and any internal numbers in the un-provided 05_Hardware doc and AutoEat_Cost_Sheet.xlsx; and the specific compute claims in the un-provided plan document. These are institution- and document-specific and must be confirmed against the missing files.